higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper)
NQF LEVEL 2

21 FEBRUARY 2019

This marking guideline consists of 10 pages.

Copyright reserved

✓ = 1 MARK	√ = ½ MARK
Accept all mathematically valid solutions.	

QUESTION 1

1.1
1.1.1 A
1.1.2 B
1.1.3 C
1.1.4 A
1.1.5 B

(5 × 1) (5)

1.2 1.2.1

$$\frac{125}{100}=\frac{5}{4}\checkmark$$

(1)

1.2.2 Let $x = 2,94343.....(1)$

$(1)\times 10:\ 10x = 29,4343.....(2)$ √

$(2)\times 100 : 1000x = 2943,4343.....(3)$ √

$(3)-(2)$

$$990x = 2914 \ \surd$$

$$\therefore x = \frac{2914}{990} \ \surd$$

$$= \frac{1457}{495} \ \checkmark$$

(3)

1.3

$$\frac{7\sqrt{108}-4\sqrt{75}-3\sqrt{48}}{-\sqrt{27}}$$

$$=\frac{7\sqrt{36\times 3}-4\sqrt{25\times 3}-3\sqrt{16\times 3}}{-\sqrt{9\times 3}} \ \checkmark$$

$$=\frac{42\sqrt{3}-20\sqrt{3}-12\sqrt{3}}{-3\sqrt{3}} \ \checkmark$$

$$=\frac{10\sqrt{3}}{-3\sqrt{3}} \ \checkmark$$

$$=-\frac{10}{3} \ \checkmark$$

$$=-3\frac{1}{3}$$

(4)

1.4 1.4.1 $5(x+y)^0 = 5$ ✓ (1)

1.4.2
$$\frac{2x^2y \times 4xy^{-3}}{8xy^{-2}}$$
$$= \frac{8x^3y^{-2}}{8xy^{-2}} \checkmark$$
$$= x^2 \checkmark$$ (2)

1.4.3
$$\frac{(49a^{-6})^{-\frac{1}{2}}}{\left(\sqrt{81b^{-6}}\right)^{-1}}$$
$$= \frac{\sqrt{81b^{-6}}}{\sqrt{49a^{-6}}} \quad \surd$$
$$= \frac{\sqrt{81}.\sqrt{b^{-6}}}{\sqrt{49}.\sqrt{a^{-6}}} \checkmark$$
$$= \frac{9b^{-3}}{7a^{-3}} \checkmark$$
$$= \frac{9a^3}{7b^3} \quad \surd$$

Alternative 1
$$\frac{(49a^{-6})^{-\frac{1}{2}}}{\left(\sqrt{81b^{-6}}\right)^{-1}}$$
$$= \frac{49^{-\frac{1}{2}}a^3}{\left(\sqrt{81}\right)^{-1}\left(\sqrt{b^{-6}}\right)^{-1}} \quad \surd$$
$$= \frac{(9)^1 a^3}{49^{\frac{1}{2}}\left(\sqrt{b^{-6}}\right)^{-1}} \checkmark$$
$$= \frac{9\ a^3}{7\left(b^{-3}\right)^{-1}} \checkmark$$
$$= \frac{9a^3}{7b^3} \quad \surd$$

Alternate 2
$$\left(\frac{(49a^{-6})^{\frac{1}{2}}}{\sqrt{81b^{-6}}}\right)^{-1}$$
$$= \frac{\sqrt{81b^{-6}}}{(49a^{-6})^{\frac{1}{2}}} \quad \surd$$
$$= \frac{\sqrt{81}\sqrt{b^{-6}}}{\left(49^{\frac{1}{2}}a^{\frac{-6}{2}}\right)} \checkmark$$
$$= \frac{9b^{-3}}{7a^{-3}} \checkmark$$
$$= \frac{9a^3}{7b^3} \quad \surd$$

(3)

1.5 1.5.1

$$A_t = A_0\left(1+\frac{r}{100}\right)^n$$

$$\frac{A_t}{A_0} = \left(1+\frac{r}{100}\right)^n \quad \surd$$

$$\sqrt[n]{\frac{A_t}{A_0}} = 1+\frac{r}{100} \quad \surd$$

$$\sqrt[n]{\frac{A_t}{A_0}} - 1 = \frac{r}{100} \quad \surd$$

$$r = 100\left(\sqrt[n]{\frac{A_t}{A_0}} - 1\right) \quad \surd$$

1.5.2

$$4\,200 = 3\,000\left(1+\frac{r}{100}\right)^4 \quad \surd$$

$$\frac{4\,200}{3\,000} = \left(1+\frac{r}{100}\right)^4$$

$$\sqrt[4]{1,4} - 1 = \frac{r}{100} \quad \surd$$

$$0,087757 = \frac{r}{100} \quad \surd$$

$$r = 8,7757 = 8,78 \surd$$

Alternative:

$$r = 100\left(\sqrt[4]{\frac{4\,200}{3\,000}} - 1\right) = 8,78 \checkmark$$

(2 × 2) (4)

1.6 1.6.1 $a = 103, d = -4, n = 5$

Substitute in

$$T_5 = a + (n-1)d$$

$$= 103 + (5-1)(-4) \checkmark$$

$$= 103 + (5-1)(-4)$$

$$= 103 - 16$$

$$= 87 \checkmark$$

Alternative:

$$T_5 = 103 - 4 - 4 - 4 - 4 = 87 \quad \checkmark \quad \checkmark$$

(2)

1.6.2 $a = 103, d = -4, T_n = 3$

Substitute in

$T_n = a + (n-1)d$

$3 = 103 + (n-1)(-4)$ ✓

$-100 = -4n + 4$ ✓

$4n = 104$

$n = 26$ ✓ (3)

1.6.3 $a = 103, d = -4,$

Substitute in

$T_{28} = a + (n-1)d$

$= 103 + (28-1)(-4)$

$= 103 + (27)(-4)$

$= -5$ ✓

Alternative:

$3 - 4 - 4 = -5$ ✓

On day 28 (two days after the last day) there will be no water left. The answer is negative, which indicates that the water would have run out by this time. ✓ Accept any other appropriate explanation (2)

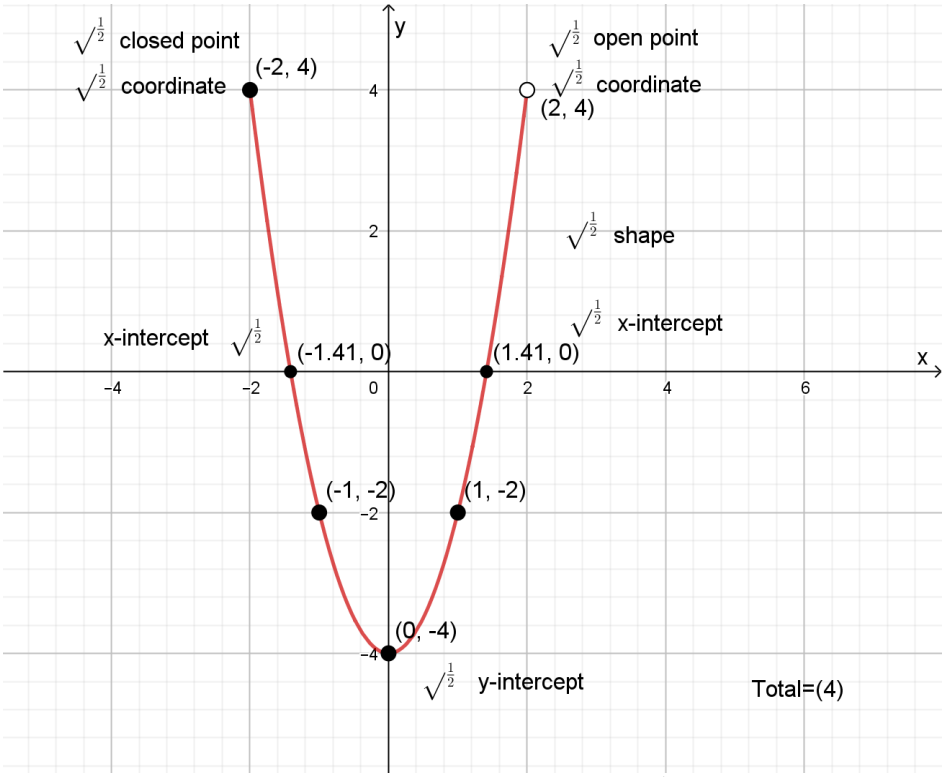
[30]

QUESTION 2

2.1 2.1.1 A
2.1.2 C
2.1.3 B
2.1.4 C
2.1.5 B
2.1.6 D

(6 × 1) (6)

2.2 2.2.1

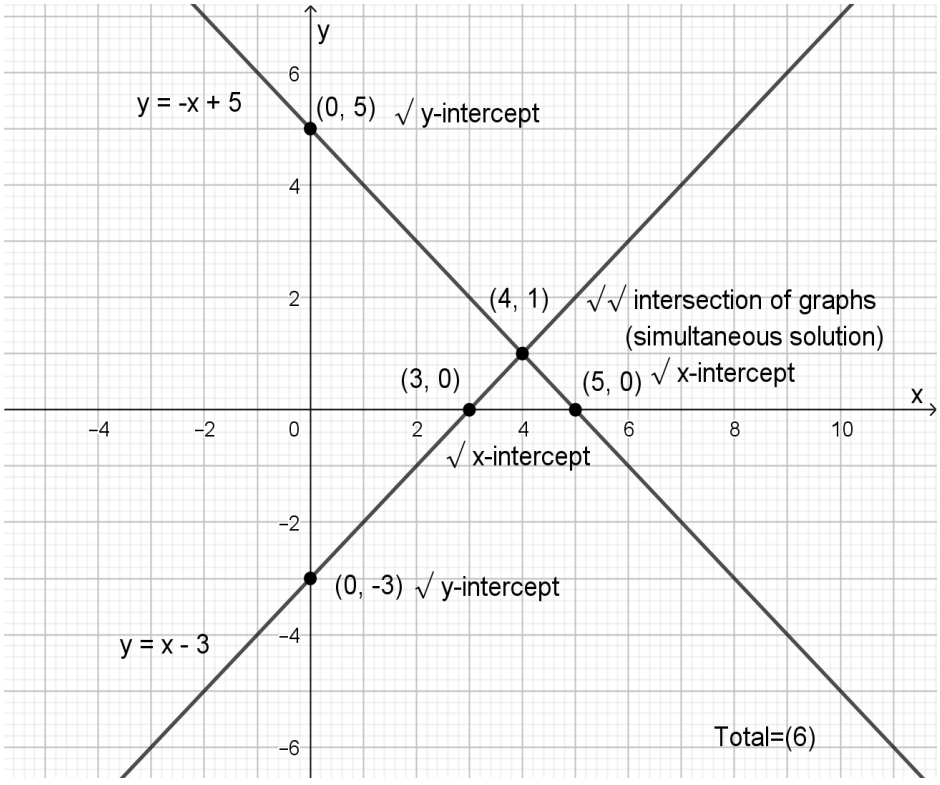


(4)

(For the sake of clarity for marking on the graph $\sqrt{\ }\frac{1}{2} = half\ \ a\ \ mark$)

2.2.2 The graph will have a maximum turning point. ✓
The graph will be inverted. (1)

2.3.1 & 2.3.2



(6)

2.4 2.4.1

$q = 2$

$y = a.b^{x} + 2$ ✓

$At\ (0, 4)$

$4 = a.b^{0} + 2$

$\therefore a = 2$ ✓

$At\ (1; 6)$

$6 = 2.b^{1} + 2$

$\therefore b = 2$ ✓ (3)

2.4.2 Exponential graph ✓ (1)

2.4.3 $y \in (2; \infty)$ ✓, $y \in \mathbb{R}$ ✓

$Alternative: \{y \mid y > 2; y \in \mathbb{R}\}$ (2)

2.4.4 $g(x)$ is a function. (1)

2.4.5
- For every x value there is only one y value
- x-values do not repeat
- Vertical line test cut once (Any 1 × 1) (1)

[25]

QUESTION 3

3.1 3.1.1

$(2x-3)(x^2 - x + 1)$

$= 2x^3 - 2x^2 + 2x - 3x^2 + 3x - 3$ √ √ √ √

$= 2x^3 - 5x^2 + 5x - 3$

3.1.2

$2x - 4x(2x-5) - (3x+21)$

$= 2x - 8x^2 + 20x - 3x - 21$ √

$= -8x^2 + 19x - 21$ √ √ √

(2 × 2) (4)

(For the sake of clarity for marking each tick represents a full mark)

3.2 3.2.1 $a^3 - ab^2$

$= a(a^2 - b^2)$ ✓

$= a(a-b)(a+b)$ ✓ (2)

3.2.2 $12x^2 - 20x + 7$

$(6x - 7)$ ✓ $(2x - 1)$ ✓ (2)

3.2.3 $25p^2 - 16 + (5p+4)(p-2)$

$= (5p-4)(5p+4) + (5p+4)(p-2)$ ✓

$= (5p+4)(5p-4+p-2)$ ✓

$= (5p+4)(6p-6)$ √

$= 6(p-1)(5p+4)$ √

Alternative:

$25p^2 - 16 + (5p+4)(p-2)$

$= 25p^2 - 16 + 5p^2 - 6p - 8$ ✓

$= 30p^2 - 6p - 24$ √

$= (5p+4)(6p-6)$ ✓

$= 6(p-1)(5p+4)$ √ (3)

3.3 $\dfrac{7xy - 21x^2y^2 + 21x^3y^3}{7xy}$

$= \dfrac{7xy(1 - 3xy + 3x^2y^2)}{7xy}$ ✓

$= 1 - 3xy + 3x^2y^2$ ✓ (2)

3.4 3.4.1 $\dfrac{1}{4} = \dfrac{x-3}{8}$

$4(x-3) = 8$ √

$4x - 12 = 8$ √

$4x = 20$ √

$\therefore x = 5$ √

3.4.2 $(x-2)(x+2) = 3x^2 + x - 10$

$x^2 - 4 = 3x^2 + x - 10$ √

$\therefore 2x^2 + x - 6 = 0$ √

$\therefore (2x-3)(x+2) = 0$ √

$\therefore x = \dfrac{3}{2}$ or $x = -2$ √

3.4.3 $9^{2x+3} = 27^{x+5}$
$(3^2)^{2x+3} = (3^3)^{x+5}$ √
$3^{4x+6} = 3^{3x+15}$ √
$\therefore 4x+6 = 3x+15$ √
$\therefore x = 9$ √

(3 × 2) (6)

3.5 3.5.1 $-2(x+3) > 5x - 76$
$-7x > -70$ √
$\therefore 7x < 70$ √
$\therefore x < 10$ √ (2)

3.5.2 Interval notation
$x \in (-\infty;10)$ ✓ where $x \in \square$ (1)

3.6 Elimination :
$3x + 4y = 4$ .....(1)
$6x - y = 26$ .....(2)
$(2) \times 4 : 24x - 4y = 104$ ....(3) √
$(1) + (3)$
$27x = 108$ √
$x = 4$ ✓
$y = 6(4) - 26$
$= 24 - 26$
$y = -2$ ✓

Alternative: Substitution
$3x + 4y = 4$ .....(1)
$6x - y = 26$.....(2)
From (2)
$y = 6x - 26$ substitute in (1) √
$3x + 4(6x - 26) = 4$ √
$\therefore 27x - 104 = 4$
$\therefore 27x = 108$
$\therefore x = 4$ ✓
$\therefore y = 6(4) - 26$
$= -2$ ✓

(3)
[25]

QUESTION 4

4.1 4.1.1 Credit card

4.1.2 Fixed expenses

4.1.3 Mashonisa/loan shark/money lender

4.1.4 Pension fund

4.1.5 Variance

(5 × 1) (5)

4.2 4.2.1 $A = -R\,111{,}23$

4.2.2 • October – December ✓ or

- The last 3 months of the year (Any 1 × 1)

4.2.3 Salary✓

4.2.4 Compound interest (there is not a constant difference, therefore compound)✓

(4 × 1) (4)

4.3 4.3.1

$A = P(1+rt)$

$A = 20\,000(1+\frac{12}{100}.8)$ ✓

$A = R39\ 200$ ✓

or

$I = \Pr t$

$I = 20\,000 \times \frac{12}{100} \times 8 = R19\ 200$ ✓

$A = R20\,000 + R19\,200 = R39\,200$ ✓

4.3.2

$A = P\left(1+r\right)^n$

$A = 20\,000\left(1+\frac{12}{100}\right)^8$ ✓

$A = R49\,519,26$ ✓

(2 × 2) (4)

4.4 4.4.1

$A_t = A_0(1+i)^n$

$R12,99 = A_0(1+\frac{9,17}{100})^{50}$ ✓

$A_0 = \frac{R12,99}{\left(1+\frac{9,17}{100}\right)^{50}}$ ✓

$A_0 = R0,16 = 16c$ ✓

4.4.2

$\text{Grand mother}: \frac{0,16}{2,00}^{\surd} \times 100 = 8\%$ ✓

$\text{You}: \frac{12,99}{160,00}^{\surd} \times 100 = 8,12\%$ ✓

(2 × 3) (6)

4.4.3 Bread relative to the salaries was less expensive 50 years ago. (1)

[20]

TOTAL: 100

Copyright reserved